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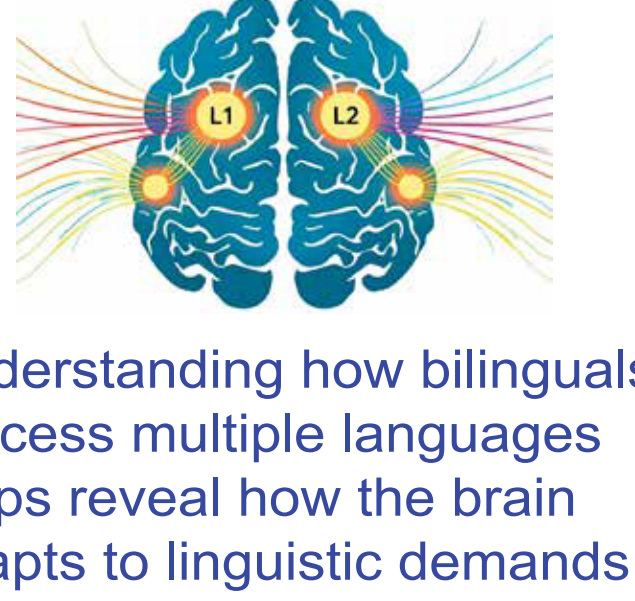
Goal: To examine how bilinguals’ brain networks **dynamically reorganize during semantic processing** in their native (L1: Spanish) vs. second (L2: Basque) language, and how these dynamics differ between **advanced** and **intermediate** speakers.

Core Question: How do large-scale brain networks adapt when bilinguals switch between their first and second language, and how does this reorganization depend on language proficiency?

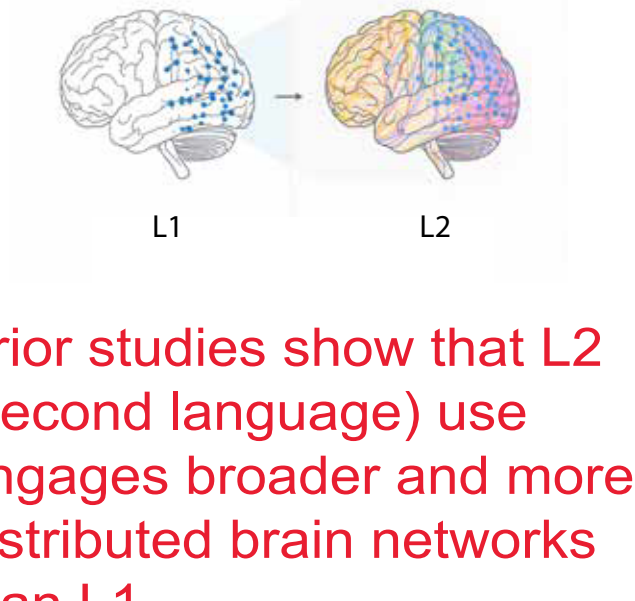
Rationale:

- Traditional fMRI studies often treat brain connectivity as static, overlooking how network interactions fluctuate over time.
- In this study, we used dynamic functional connectivity (dFC) and Hidden Markov Models (HMMs) to capture these moment-to-moment changes in network organization.
- Our work tests the Neurocognitive Adaptation Hypothesis, which predicts that greater bilingual proficiency leads to enhanced network stability, efficiency, and flexibility during second-language processing.

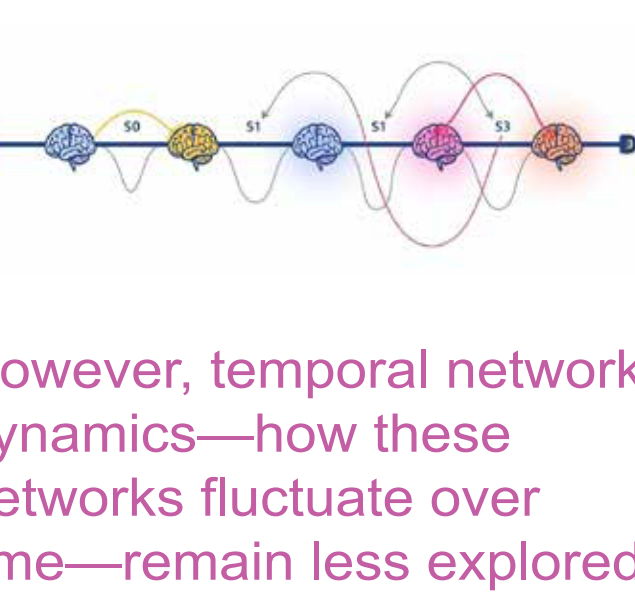
Background



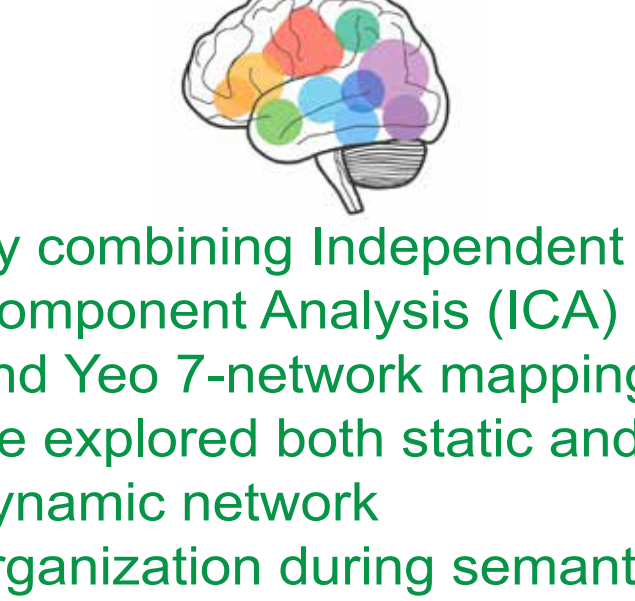
Understanding how bilinguals process multiple languages helps reveal how the brain adapts to linguistic demands.



Prior studies show that L2 (second language) use engages broader and more distributed brain networks than L1.



However, temporal network dynamics—how these networks fluctuate over time—remain less explored.



By combining Independent Component Analysis (ICA) and Yeo 7-network mapping, we explored both static and dynamic network organization during semantic processing in bilinguals.

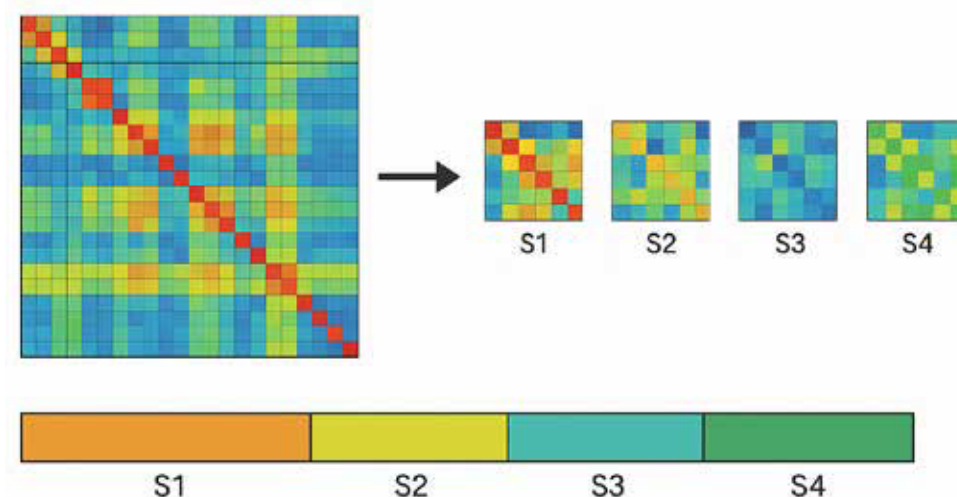
- Dynamic Functional Connectivity (dFC) captures how brain network interactions fluctuate over time.
 - This approach allows us to measure:
 - Mean Dwell Time (MDT):** how long the brain maintains a network configuration.
 - Transitions:** how often it switches between configurations.
 - Fractional Occupancy (FO):** how much time is spent in each configuration.
- Why Independent Component Analysis (ICA)?**
- We used ICA to decompose whole-brain fMRI data into spatially independent functional networks, providing a data-driven view of large-scale systems engaged during bilingual language processing.
 - This approach captures individual and group-level network dynamics without imposing predefined anatomical constraints, enabling us to track how these networks reconfigure over time across languages and proficiency levels.

- Our focus:**
- Identify core networks engaged during L1 and L2 processing.
 - Examine temporal flexibility and network occupancy across proficiency levels.
 - Relate dynamic metrics (Fractional Occupancy, Dwell Time) to bilingual proficiency.

Approach

Dynamic State Modeling and Network Integration

Time-varying functional connectivity (FC) matrices were computed and clustered. A Hidden Markov Model (HMM) cross-validated these patterns, yielding a discrete sequence of recurring “brain states.”



Network Mapping (Yeo-7 Integration)

Each ICA component was mapped to canonical Yeo-7 networks (Visual, Somatomotor, Dorsal Attention, Ventral Attention, Limbic, Control, DMN). This established a link between ICA-derived spatial maps and known functional systems.

Network Similarity $L1, L2 = corr(ZL1, ZL2)$

Methods

26 Right-handed Native Spanish speakers learning Basque

Participants



Intermediate
(40 < CEFR scores < 60, n=10)



Advanced
(70 < CEFR scores < 100, n=16)

All living in Spanish-Basque bilingual environments

Semantic Animacy Judgment Task



Reading condition – printed words shown for 1000 ms



Listening condition – auditory words (~585 mc)

Decide: Living or Non-living?

Respond with right-hand button press

 **dog** → LIVING

 **chair** → NON-LIVING

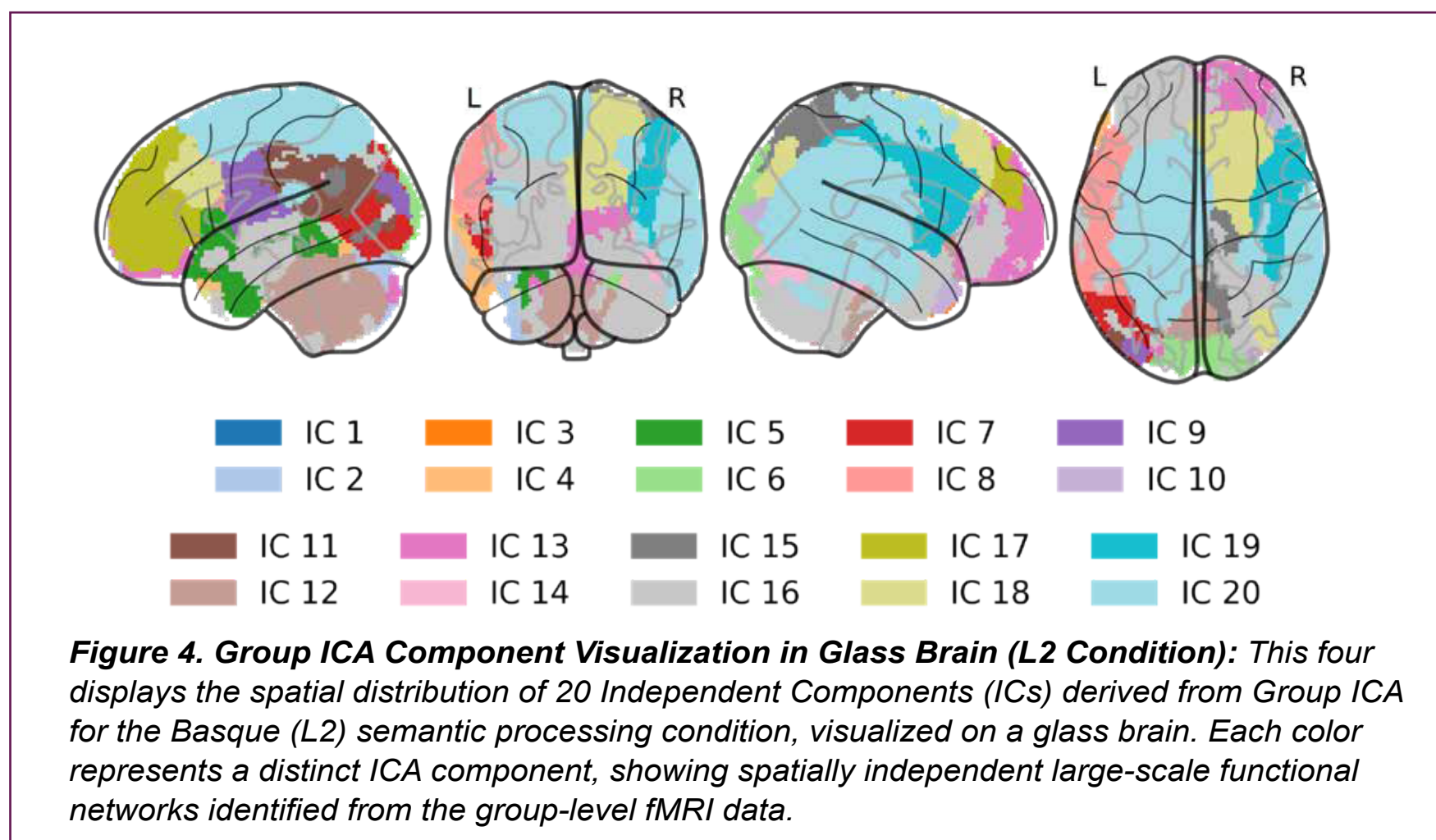
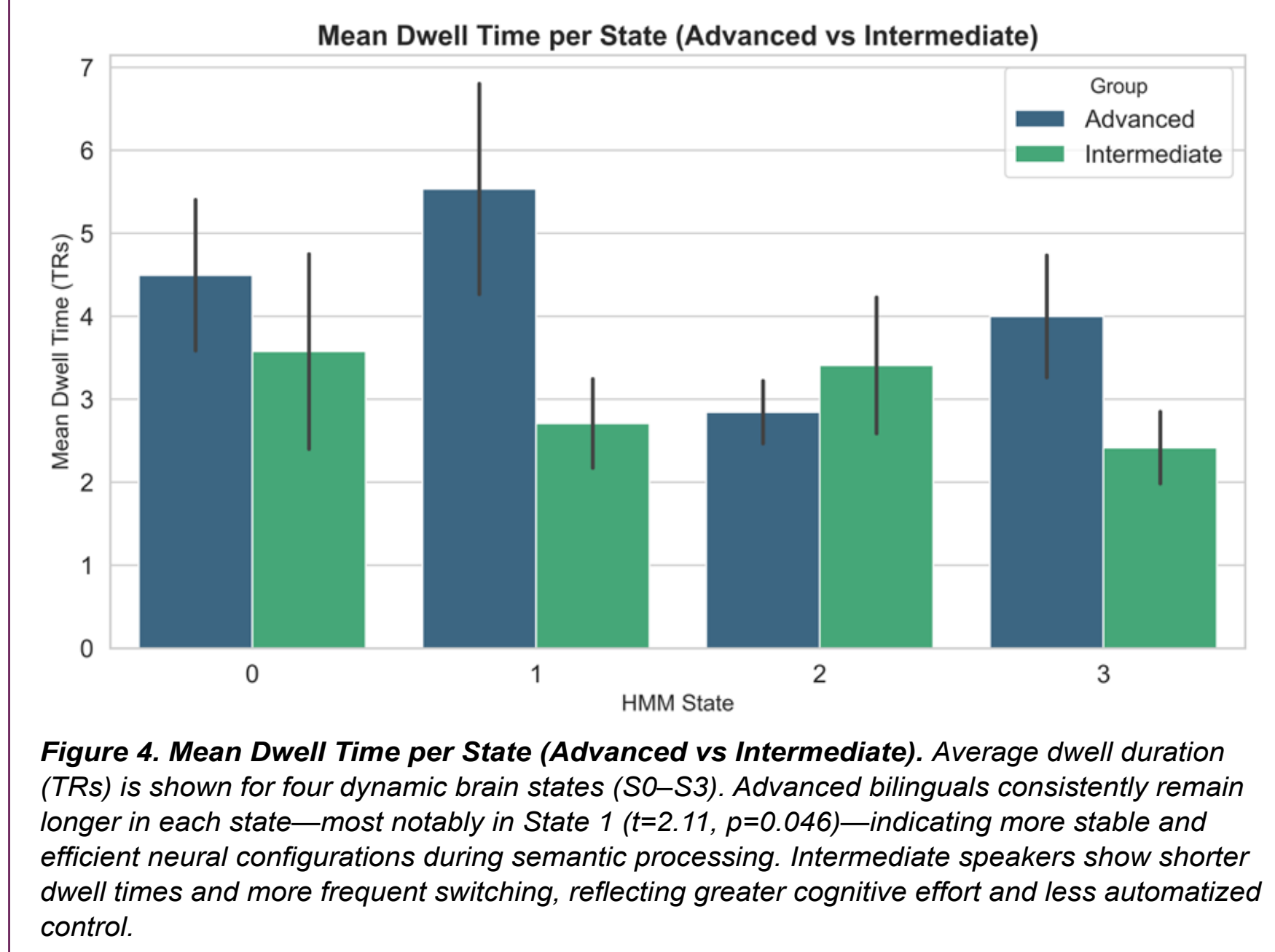
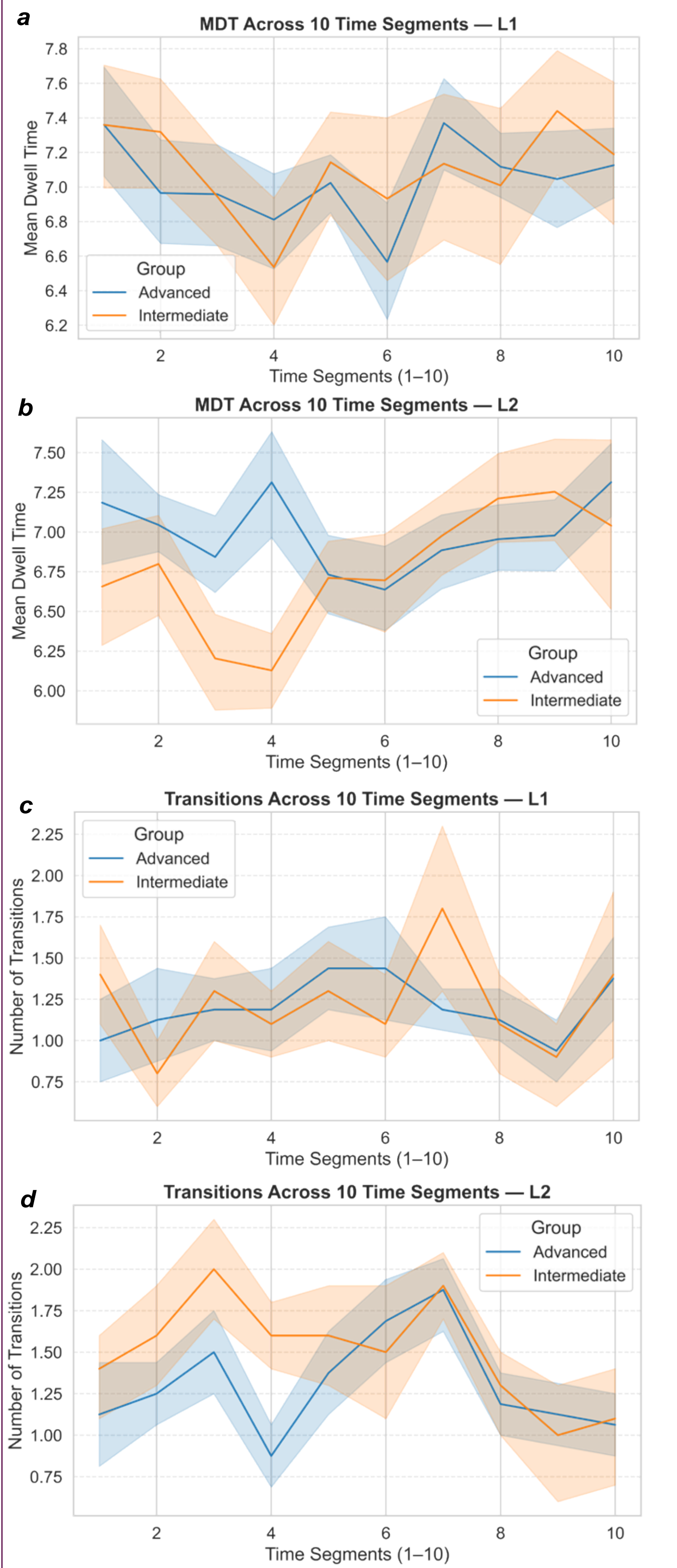
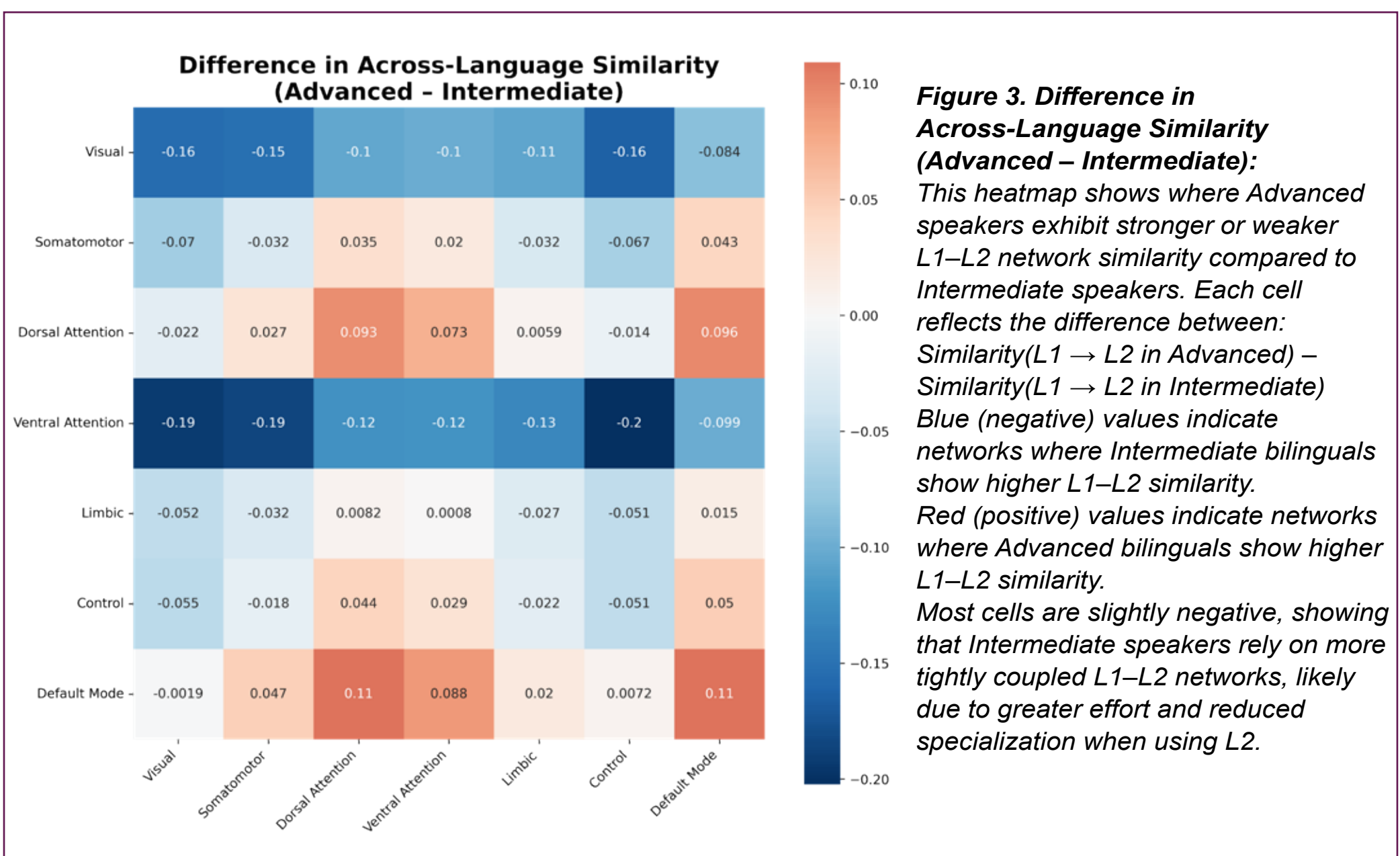
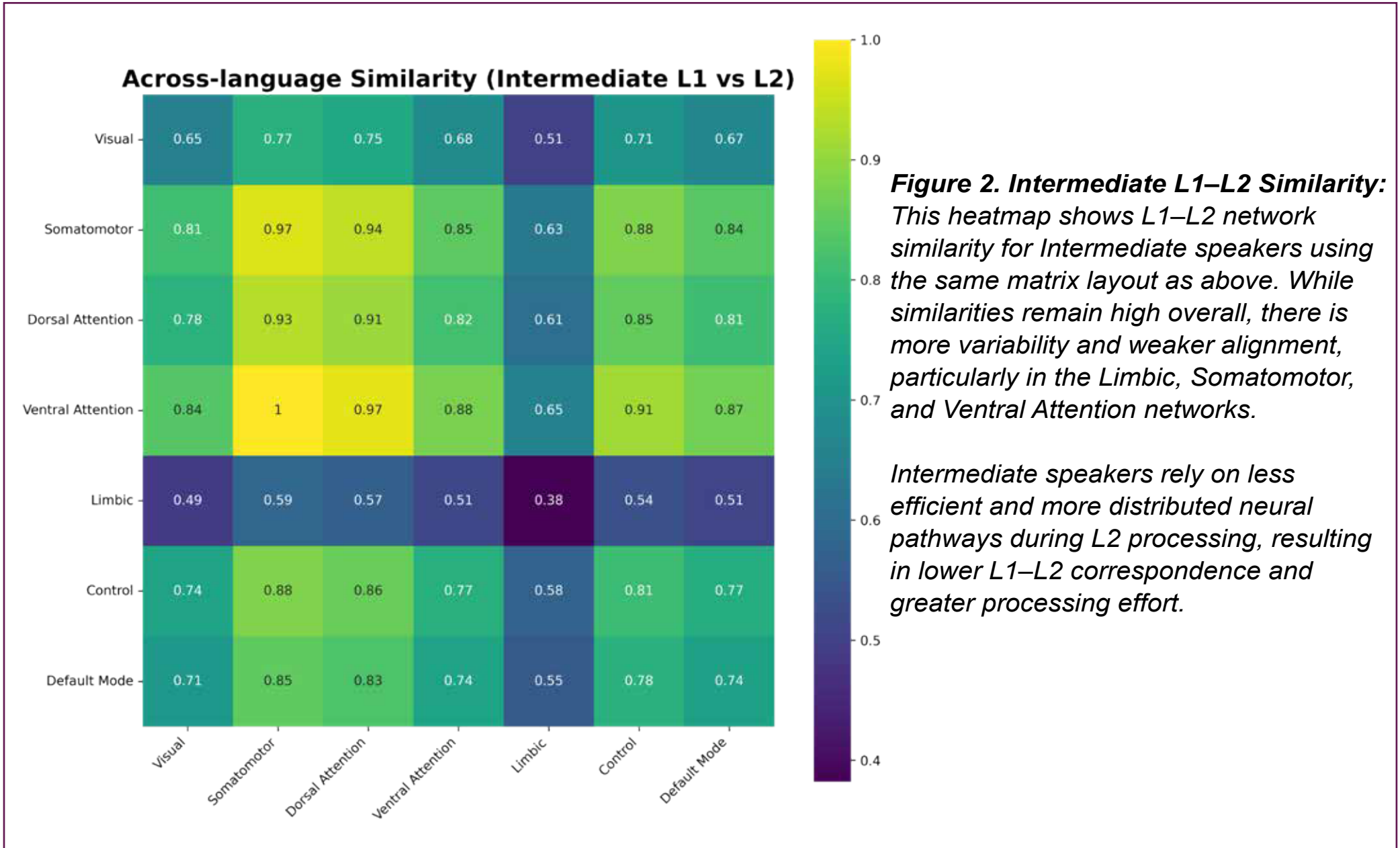
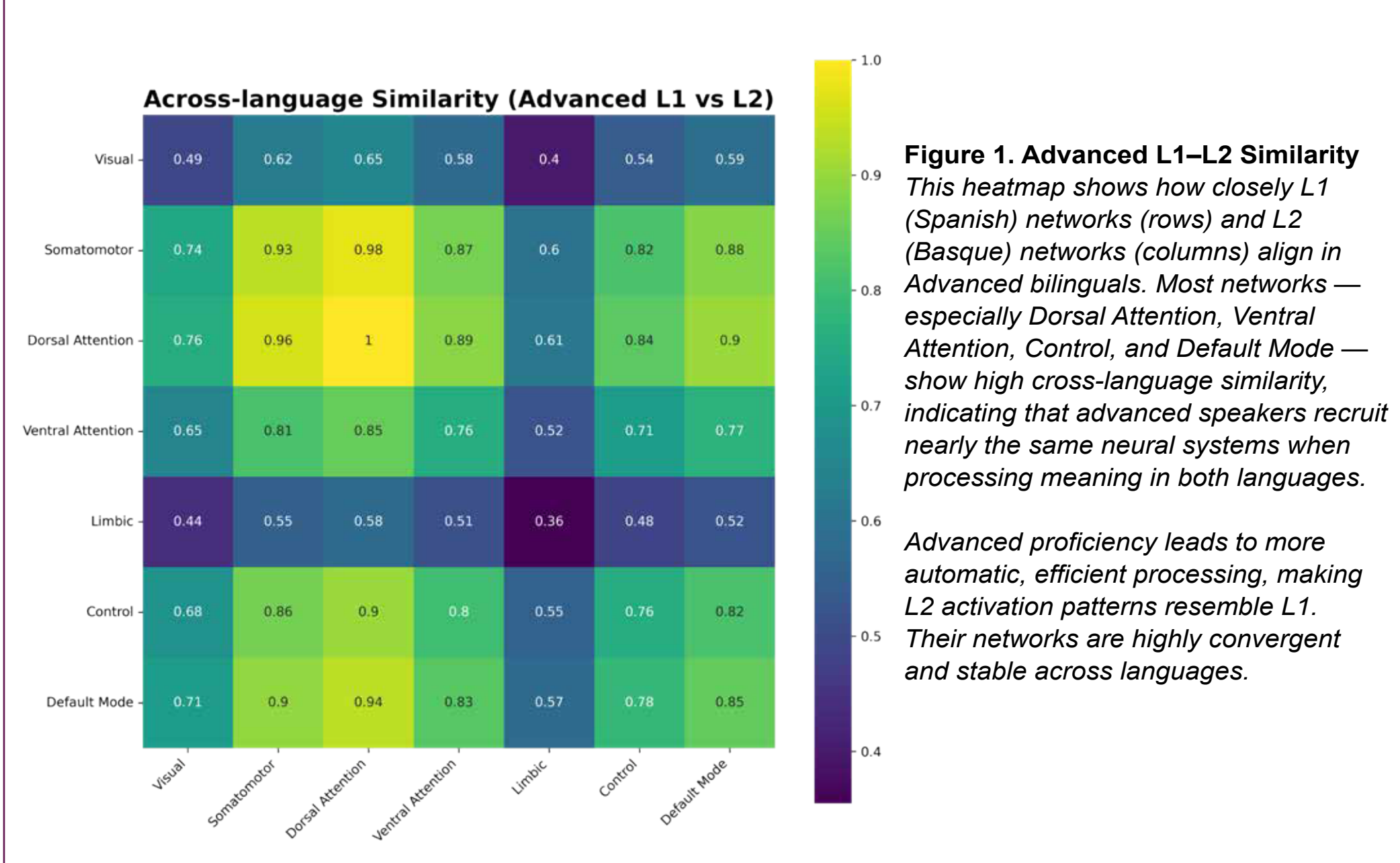
Experimental Structure (Inside the Scanner)

Action	Rest	Intermixed (auditory/visual) Stimuli	Rest	Intermixed (auditory/visual) Stimuli	Rest
Time (sec)	20	-150	20	-150	20

Language 1 (Spanish 🇪🇸) — 3 runs — 48 stimuli/run

Language 2 (Basque 🇧🇪) — 3 runs — 48 stimuli/run

Order counterbalanced across participants to avoid switching effects.



Summary of findings

Our results show that cross-language neural similarity increases with proficiency. Advanced bilinguals rely on highly overlapping networks for L1 and L2, indicating automated and efficient semantic processing across languages. Intermediate speakers show more divergent L1–L2 patterns, reflecting greater effort, reduced automaticity, and heavier reliance on perceptual and control systems in L2.

Cross-language correlation maps reveal that intermediates show broader, less specialized coupling, whereas advanced speakers engage more selective and efficient attention and control networks. Dynamic analyses further show that advanced bilinguals maintain longer dwell times and fewer transitions, supporting greater temporal stability during L2 processing.

Overall, bilingual proficiency shapes how flexibly and efficiently large-scale brain networks interact, revealing a progressive adaptation of neural architecture as second-language skills strengthen.

References

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